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Lunar soils, simulants and lunar construction materials: An overview

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Abstract

One of the hot subjects in space civil engineering is the research for establishing outposts on space bodies, starting with the Moon. This wide research topic mainly covers learning about ground conditions and soils of these extraterrestrial bodies, looking for means of using the soil in producing construction materials, designing structures appropriate for local conditions, and finally choosing construction techniques to be applied using mainly these materials. In this paper, as the title indicates, an overview is presented on three aspects of using local materials in lunar construction activities: nature of in-situ soils obtained with lunar missions, the variety of lunar soil simulants produced on Earth, and their use in producing lunar construction materials.
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1. Introduction

Engineering, and particularly civil engineering is defined in 1828 by Tredgold (Toklu, 2000) as "... the art of directing great sources of power in Nature for the use and convenience of man, as the means of production and of traffic in States both for external and internal trade, as applied in the construction of roads, bridges, aqueducts, canals, river navigation and docks, for internal intercourse and exchange; and in the construction of ports, harbours, moles, breakwaters and lighthouses, and in the art of navigation by artificial power for the purpose of commerce; and in the construction and adaptation of machinery; and in the drainage of cities and towns." This definition, forwarded for terrestrial constructions, will be verified for extraterrestrial constructions, starting with works on the Moon. Indeed, construction materials to be used in lunar structures will be mostly raw materials found on the Moon or materials manufactured using lunar soil as the main input. The main reason for this is the high cost of transporting materials from Earth to the Moon (Toklu, 2004; Toklu, 2005).

The only in-situ material to be found readily available on the Moon is the regolith, i.e. soil on the surface of the Moon with depth changing from a few centimeters to some tens of meter. This is the material that could be used as is or after being subject to operations to yield construction materials like lunar cement, lunar concrete, lunar brick or the like. The operations to obtain original lunar samples available for research is currently less than 400 kg, of which 382 kgs are brought through US Apollo missions, and about 300 g through Soviet Luna missions. China made an addition about 2 kg to this amount very recently (Ellery, 2021). Although it is certain that there will be continuous additions to these samples in the future, the quantity at hand will never be sufficient for all studies and for all the researches to be conducted on Earth.

Due to the limited amount of lunar soil and rock samples, researchers on Earth, in many countries, launched

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researches on producing soils having basic characteristics similar to lunar soil samples, i.e. simulants on which experiments can be carried on as if on real lunar soils. Some of these researches are further focused on manufacturing materials that can be used on the Moon as cement, concrete, brick etc.

This paper is arranged to give a summary of the works cited above. Section 2 gives a brief account of missions that brought lunar soil samples. In Section 3, studies on manufacturing lunar soil simulants are reviewed. Construction materials that are forwarded using these simulants are overviewed in Section 4. The last section, Section 5, is intended to give some concluding remarks about all these works.

2. Lunar missions and lunar soil samples

The Luna Program was a long-duration lunar exploration program conducted by the Soviet Union between 1959 and 1976, to collect information about the Moon and its environment. This program enabled bringing to Earth three sets of lunar soil samples through Luna 16 (in 1970), Luna 20 (in 1972), and Luna 24 (in 1976) missions. Remote analyses of lunar soil through all Luna sessions and laboratory investigations of the three sets of samples transported to Earth are discussed elsewhere (Cherkasov and Shvarev, 1973, 1977; Slyuta, 2014).

There have been six Apollo missions at which lunar soil samples are brought to the Earth (Refer Table 1). Landing sites of Luna and Apollo missions can be seen at the map provided by NASA (https://nssdc.gsfc.nasa.gov/planetary/lunar/moon_landing_map.jpg).

These missions are so arranged that Apollo 11 and 12 landed on mare areas, Apollo 14 and 15 landed on highlands, and Apollo 16 and 17 landed on highland – mare contact areas (Heiken et al, 1991). "Mare" areas on the Moon are known to be the relatively darker areas that can be easily spotted with naked eye. Mare areas are plains that are located in lower altitudes compared to the "highlands". Mare areas are formed from solidifying basaltic lavas. Highlands, on the other hand, are remarkably lighter in color and are known to consist of anorthosites, which are a type of igneous rocks formed from lavas cooling down slowly (Mckay and Ming, 1990). Ivanov et al. (1973) presented an extensive comparison of the characteristics of lunar *mare* and *highland* regolith samples brought to Earth by Luna Missions. Their results primarily indicated that lunar *highland regolith* samples exhibited: increased amount of coarse-grained particles with relatively higher contents of original rock fragments, being majorly composed of primary igneous and metamorphic rocks over slags, breccias and sinters, yielding slightly lower density and also having much lighter color, when they are compared to lunar mare regolith samples (Ivanov et al., 1973).

Scientific world has sufficient information about properties of lunar soils brought to Earth, especially on the ones brought through Apollo missions (Heiken et al., 1991; Hill et al., 2007; McKay et al., 1991; Robens et al., 2008; Zarzycki and Katzer, 2019). Chemical composition of lunar soils is tabulated in Table 2 (Zarzycki and Katzer, 2019). The data on this table is visualized in Fig. 1 where it can be clearly seen that lunar soils shows important variations as to the locality they are taken.

In fact, there are trace elements in these soils in addition to the components listed in this table. For instance, in the sample 15,601 brought in Apollo 15 mission, the trace elements are found, in parts per million as, 35.1 Scandium, 200 Vanadium, 48.9 Copernicium, 135 Barium, 11.3 Lanthanum, 29 Cerium, 6.3 Samarium, 1.01 Europium, 1.33 Terbium, 9.7 Dysprosium, 5.2 Ytterbium, 0.9 Lutetium, 4.9 Hafnium, 0.60 Tantalum, 1.52 Thorium, 0.46 Uranium, and a plausible value of 170 Nickel (Heiken et al., 1991).

In 2020, China has become the third country bringing lunar regolith samples with the successful delivery of Chang'e 5, China's firts lunar sample-return mission, bringing around 2 kg regolith for the first time after the last Luna 24 mission made by Soviet Union in 1976. Chang'e 5 is the first mission ever landing on the basaltic area in the northeastestern Oceanus Procellaum in the northwest of the Moon and it was detected that the sampling site had a significantly younger basalt age (compared to the sites of Apollo and Luna missions) and hence this sampling site has been assessed to have high scientific significance by reserachers (Zhang et al., 2022). The studies made on the regolith samples brought by Chang'e 5 have revealed that it contains basaltic minerals that are dominated by pyroxene, as well as agglutinates containing minerals and glasses, glass beads which contain native Fe, where the diversified debris fragments were with approximately 44.5% pyroxene, 30.4% plagioclase, 3.6% olivine, 6.0% ilmenite, and 15.5% glass (Zhang et al., 2022).

The compositional information of lunar regolith is critical for the task of simulating soil samples that would be employed in engineering experiments aiming to provide

Table 1
Apollo Lunar Landings and returned samples (Adapted from: https://nssdc.gsfc.nasa.gov/planetary/lunar/apolloland.html).

Apollo Mission	Launch Date	Landing Site	Sample Return (kg)
11	16 Jul 1969	Mare Tranquilitatis	21.55
12	14 Nov 1969	Oceanus Procellarum	34.35
14	31 Jan 1971	Fra Mauro	42.28
15	26 Jul 1971	Hadley Rille	77.31
16	16 Apr 1972	Descartes	95.71
17	07 Dec 1972	Taurus-Littrow	110.52

Table 2
Chemical compositions of lunar soils brought through Apollo missions *(Weight %) (LOI: Loss in ignition)*.

Mission	Specimen	SiO_2	TiO_2	Al_2O_3	Cr_2O_3	Fe_2O_3	FeO	MnO	MgO	CaO	Na_2O	K_2O	P_2O_5	S	LOI
Apollo 11	10,084	42.1	7.8	13.7	0.3	—	15.8	0.2	7.9	12	0.5	0.1	0.1	—	—
	—	42.2	7.8	13.6	—	—	15.3	0.2	7.8	11.9	0.47	0.16	0.05	—	0.12
Apollo 12	12,030	46.6	3.6	14.2	0.35	—	15.4	0.22	9.7	10.4	0.43	0.24	—	—	—
	12,001	46	2.8	12.5	0.41	—	17.2	0.22	9.7	10.9	0.48	0.24	—	—	—
Apollo 14	14,259	48.2	1.73	17.6	0.26	—	10.41	0.14	9.26	11.25	0.61	0.51	0.53	—	—
	14,163	47.3	1.6	17.8	0.2	—	10.5	0.1	9.6	11.4	0.7	0.6	—	—	—
	Average	48.1	1.7	17.4	—	—	10.4	0.14	9.4	10.7	0.7	0.55	0.51	—	—
Apollo 15	15,071	46.95	1.6	12.7	0.465	—	16.29	0.217	10.75	10.49	0.33	0.092	0.16	0.07	—
Apollo 16	61,221	45.35	0.49	28.25	—	—	4.55	0.06	5.02	16.21	0.42	0.09	0.1	0.06	—
	61,141	45.2	0.58	26.4	0.16	—	5.29	0.7	6.1	15.32	0.52	0.14	0.12	—	—
	62,241	44.65	0.56	27	—	—	5.49	0.7	5.84	15.95	0.44	0.13	0.1	—	—
	64,801	44.9	0.47	27.7	0.11	—	5.01	—	5.69	15.7	0.51	0.22	0.16	—	—
	67,461	44.77	0.37	28.99	—	—	4.35	0.07	4.2	16.85	0.44	0.06	0.05	—	—
	—	45	0.54	27.3	—	—	5.1	0.3	5.7	15.7	0.46	0.17	0.11	—	0.07
Apollo 17	79,221	41.67	6.52	13.57	0.42	—	15.37	0.21	10.22	11.18	0.34	0.09	0.06	0.12	—
	71,501	39.82	9.52	11.13	0.46	—	17.41	0.25	9.51	10.85	0.32	0.07	0.06	0.12	—
	71,061	40.09	9.32	10.7	0.49	—	17.85	0.24	9.92	10.59	0.36	0.08	0.07	0.13	—
	70,051	42.2	5.09	15.7	—	—	12.4	0.15	10.3	11.5	0.24	0.07	—	—	—

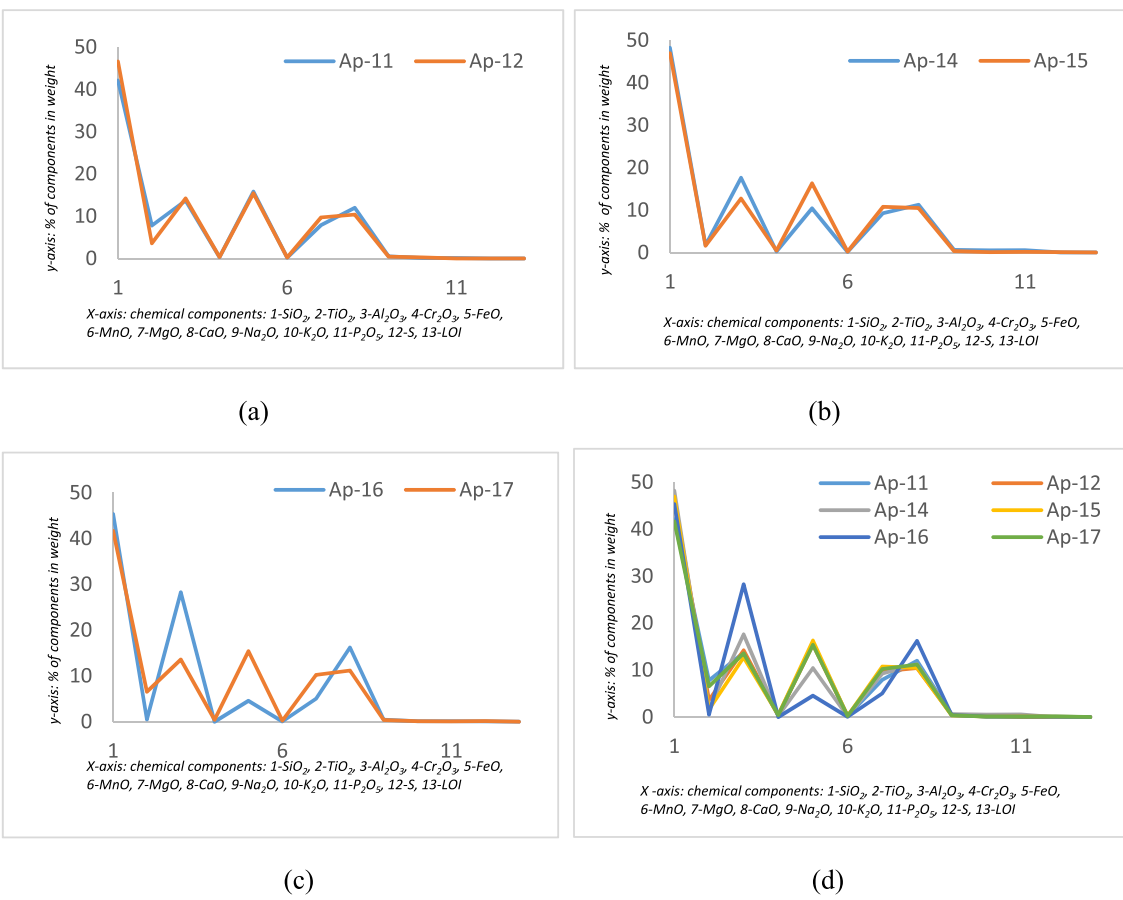

Fig. 1. Chemical compositions of lunar soils: (a) Mare areas; Apollo 11 and 12, (b) Highlands; Apollo 14 and 15, (c) Areas in between; Apollo 16 and 17, (d) All areas; x-axis is for chemical components: 1-SiO_2, 2-TiO_2, 3-Al_2O_3, 4-Cr_2O_3, 5-FeO, 6-MnO, 7-MgO, 8-CaO, 9-Na_2O, 10-K_2O, 11-P_2O_5, 12-S, 13-Loss in ignition (LOI), y-axis for percentages of components in weight.

understandings on the behavior of this primary lunar resource. For instance, lunar construction materials to be obtained out of lunar regolith are expected to be produced by subjecting the regolith mainly to some chemical processes. Hence, a realistic lunar soil simulant should be able to reflect the chemical properties of the regolith samples, in

order to provide reliable results when such experiments on manufacturing and testing construction materials is of concern.

3. Lunar soil simulants

In order to provide advancements in the success of future lunar missions and in the ultimate goal of colonization of the Moon, several experimental studies should be carried out beforehand regarding the tasks of landing and mobility on the Moon's surface, excavation and drilling practices, as well as building lunar infrastructures and habitations. In fact, simulants are needed for testing a wide variety of specific applications on the Moon. A web-based survey on 35 lunar projects has shown that simulants are needed for different kinds of applications, of which 13% is for construction and 17% is for excavation (Refer Fig. 2). All of these scientific and engineering investigations require large quantities of lunar regolith to be employed in the experiments. However, real lunar regolith samples are very limited in quantity and usually not accessible since only around 380 kg of soil and rocks were brought to Earth with the Apollo lunar missions (Lim et al., 2017). Hence, there has been a need to produce alternative materials to substitute lunar regolith needed to be employed in the scientific investigations. Lunar simulants are materials that are made from terrestrial resources to mimic lunar dusts and soils belonging to locations of interest (i.e. lunar highland or mare) which have varying chemical compositions and particle characteristics, with the purpose of being used in experimental studies (Naser, 2019).

Taylor et al. (2016) provides noteworthy information on the history of lunar simulants and reports that the need for the use of lunar simulants had started in the early 1970′s within the investigations carried out by NASA. The mentioned study also presents a list of lunar soil simulants that

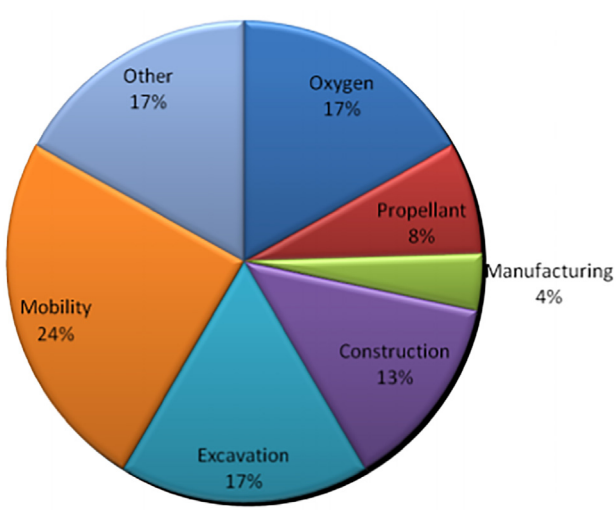

Fig. 2. Pie-Chart Summarizing Simulant Needs by Specific Applications/Processes (LEAG-CAPTEM, 2010).

is renowned within the researchers working in the field of lunar studies. This notable list, which is based on the information presented within the report (LEAG-CAPTEM, 2010) prepared by the Simulant Working Group of the Lunar Exploration Analysis Group and Curation and Analysis Planning Team for Extraterrestrial Materials (LEAG-CAPTEM SWG) and submitted to the Planetary Science Subcommittee of the NASA Advisory Council, only covers simulants that were developed until the year 2010. Some of the recent studies in related literature are observed to make use of this list of simulants, even though numerous new lunar regolith simulants of varying qualities have been developed in the last decade. Toklu and Akpinar (2019) presented a relatively extended version of this list covering information about more simulants developed later on.

Table 3 presented in this study aims to provide further information about the lunar soil simulants that were developed within the last decade, in addition to the simulants listed originally in (LEAG-CAPTEM, 2010), as well as in Taylor et al. (Taylor et al., 2016), which uses the former reference in its core. For this purpose, an extensive literature survey has been carried out covering the publications made until the end of August 2021, and as a result, 22 additional lunar soil simulants developed have been included beside the ones presented originally in (LEAG-CAPTEM, 2010). Basic information regarding the type, source materials and the intended use of these lunar soil simulants are also presented below, following Table 3.

Minnesota Lunar Simulant (MLS-1) was developed by the University of Minnesota and it was first introduced at the Second Lunar Science Conference in the year 1971. MLS-1 was produced as a lunar mare soil by using a high TiO_2-content of basalt as the source material. Later on MLS-1 was produced with varying glass contents and particle size distributions, with the intention of only mimicking high-Ti lunar regolith samples from Apollo 11 without reflecting exact mineralogical and chemical characteristics. MLS-2 is also developed by the same group with the intention of mimicking lunar highland soil samples (Taylor et al., 2016; LEAG-CAPTEM, 2010).

Arizona Lunar Simulant (ALS) was reported to include the source material of crushed rock from the Pomona basalt flow from Hanford, Washington, US. This lunar soil simulant was developed as mare type of simulant for general use purposes (Desai, 1991).

JSC-1 (Johnson Space Center) simulant is perhaps the most well-known lunar soil simulant that is widely used in previous studies. It was developed in 1993 using volcanic ash from the San Francisco volcano field near Flagstaff, Arizona, US. This simulant was developed with the intention of mimicin mare type of lunar regolith samples and was proposed for general use. When JSC-1 simulant was used up, JSC-1A was produced as a clone of the former simulant. Later on, JSC-2A which is similar to JSC-1A simulant was developed. JSC-2A is a basaltic mare simulant. Even though JSC-2A is known to be similar to

Table 3
List of lunar regolith simulants developed until August 2021.

Simulant	Country	Year	Related Reference
MLS-1 MLS-1P	USA	1971	Weiblen et al. in 1990,- as mentioned in (LEAG-CAPTEM, 2010)
MLS-2			Tucker et al. in 1992,- as mentioned in (LEAG-CAPTEM, 2010)
ALS	USA	1991	Desai et al. in 1991,-as mentioned in (LEAG-CAPTEM, 2010)
JSC-1 JSC-1A			McKay et al. in1994,-as mentioned in (LEAG-CAPTEM, 2010)
JSC-1AF	USA	1993	
JSC-2A			
FJS-1(type 1)FJS-1			Kanamori et al.in1998, as mentioned in (LEAG-CAPTEM, 2010)
(type 2)FJS-1	Japan	1998	
(type 3)			
MKS-1	USA		Carpenter in 2005,-as mentioned in (LEAG-CAPTEM, 2010)
OB-1	Canada	2009	Battler & Spray-in 2009,- as mentioned in (LEAG-CAPTEM, 2010)
CHENOBI	Canada		https://www.evcltd.com/index_005.htm as mentioned in (LEAG-CAPTEM, 2010)
CAS-1	China	2008	Zheng et al.in 2008,-as mentioned in (LEAG-CAPTEM, 2010)
GCA-1	USA	2008	Taylor et al.in 2008,- as mentioned in (LEAG-CAPTEM, 2010)
NU-LHT-1M	USA	2009	Stoeser et al. in 2009,-as mentioned in (LEAG-CAPTEM, 2010)
NU-LHT-1D			
NU-LHT-2M			
NU-LHT-2C			
Oshima	Japan	2008	Sueyoshi et al, 2008, as mentioned in (LEAG-CAPTEM, 2010)
Kohyama,	Japan	2008	Sueyoshi et al, 2008,- as mentioned in (LEAG-CAPTEM, 2010)
NAO-1	China	2009	Li et al.-in 2009,-as mentioned in (LEAG-CAPTEM, 2010)
CLRS-1	China	2009	Chinese Acad. of Sci. in 2009,- as mentioned in (LEAG-CAPTEM, 2010)
CUG-1	China	2010	He et al, 2010, as- mentioned in (LEAG-CAPTEM, 2010)
GRC-1	USA		Oravec et al, 2010, asmentioned in (LEAG-CAPTEM, 2010)
GRC-3			
TJ-1	China	2010	Jiang et al. in 2010, asmentioned in (LEAG-CAPTEM, 2010)
TJ-2			
KOHLS-1	Korea	2010	Lee et al, 2012
BP-1	USA	2010	Rahmatian and Metzger, 2010, as mentioned in (LEAG-CAPTEM, 2010)
CSM-CL	USA	2010	Dreyer & Van Susante, 2010
OPRH 2N/H2W/H3N	USA	2010	https://www.offplanetresearch.com/simulants-feedstocks-and-additives
OPRL 2N/L2W	USA	2010	
ALRS-1	Australia	2012	Bonnano & Bernold, 2015
TRI-1	India	2014	Sreenivasulu, 2014; (Jayalekshmi et al., 2019)
QH-E	China	2015	(Wei-lie et al., 2016)
EAC-1	Germany (by ESA)	2017	Nash et al., 2017Engelschiøn et al., 2020
EAC-1A	Germany	2020	
CLDS-i	China	2017	Tang et al., 2017
BHLD20	China	2017	Sun et al., 2017
NEU-1a	China	2017	Li, 2016; (Liu et al., 2017)
NEU-1b	China	2017	Li, 2016; (Liu et al., 2017)
KLS-1	Korea	2018	Byung-Hyun Ryu et al., 2018
DNA-1	Italy	2014	Cesaretti et al., 2014Marzulli & Cafaro, 2019
DNA-1A		2019	
TUBS-M	Germany	2018	Linke et al., 2018, Linke et al., 2020Linke et al., 2018, Linke et al., 2020
TUBS-T	Germany	2018	
LHS-1	USA	2019	Cannon & Britt, 2019; Britt & Cannon, 2020Cannon & Britt, 2019; Britt & Cannon, 2020
LMS-1	USA	2019	
UoM-B	UK	2020	Just et al., 2020Just et al., 2020
UoM-W	UK	2020	
LSS-ISAC-1	India	2020	Venugopal et al., 2020a; Venugopal et al., 2020b
TBG-1	Turkey	2021	Toklu et al. 2022

JSC-1A, it is reported that the different synthesis method used in its production is expected to yield certain differences in its compositional and mineralogical characteristics when compared to JSC-1A (Taylor et al., 2016; Zocca et al., 2020; LEAG-CAPTEM, 2010; Sibille et al., 2006).

FJS, which is also known as Fuji Japanese Simulant was developed in Japan by making use of Fuji Lava base as the source. Types 1 &2 are reported to be low-Ti mare type of lunar soil simulants where Type 3 was intended to be a high-Ti mare lunar soil simulant. This simulant has been

reported to be used for general use (LEAG-CAPTEM, 2010; Sueyoshi et al., 2008).

OB-1 is a lunar simulant produced by Canada with the intention to mimic highland lunar regolith samples. This simulant is reported to contain Shawmere Anorthosite and fayalitic olivine slag glass CHENOBI simulant is known to be a modified version of OB-1 simulant which contains no olivine slag but plasma-melted Shawmere Anorthosite instead. Both of these lunar simulants were developed for use in geotechnical purposes (LEAG-CAPTEM, 2010).

GCA-1 is reported to be developed by Goddard Space Center in USA. This simulant aimed to mimic low-Ti mare lunar soil samples and intended to be used for geotechnical studies (Taylor et al., 2016; LEAG-CAPTEM, 2010).

The NU-LHT series of lunar soil simulants are also known as NASA/USGS Lunar Highlands Type simulants. NU-LHT is reported to include the source materials of Stillwater Norite, Anorthosite, and Hartzburgite, and Twin Sisters Dunite. These simulants were developed for general use (Stoeser, D., Wilson S., 2010).

Oshima and Kohyama are lunar soil simulants made in Japan that aim to mimic high-Ti mare lunar regolith samples and regolith samples between highlands and mare types, respectively. The former simulant includes Oshima lava base and the latter simulant includes Kohyama gabbro base as the source. Both simulants are reported to be for general use (LEAG-CAPTEM, 2010; Sueyoshi et al., 2008).

CAS-1 CLRS-1, CUG-1 and TJ-1 are lunar simulant that were all developed by China as Low-Ti mare type. Among these, CAS-1 which was developed by Chinese Academy of Sciences, was reported to be designed to reflect mineralogical and chemical properties of Apollo 14 soil samples and was intended for general use. It is reported to be made of scoria from Jinlongdingzi Volcano in Jilin Province of China (Zheng et al., 2009). Similarly, Chinese Lunar Regolith Stimulant (CLRS) which was also reported to be developed by Chinese Academy of Sciences for general use (LEAG-CAPTEM, 2010). CUG-1 (China University of Geosciences) was made using basaltic volcanic scoria from northeast China in order to be employed in engineering studies (He et al., 2010). TJ-1 simulant developed by Tong Ji University was reported to be intended for geotechnical use (LEAG-CAPTEM, 2010).

NAO-1 is also a Chinese lunar soil simulant developed by National Astronomical Observatories of Chinese Academy of Sciences. This simulant aims to mimic lunar highland regolith and it was reported to be designed to reflect the mineralogical and chemical characteristics of Apollo 16 samples. The source material for this simulant that was developed for general use, was reported to be gabbro obtained from Yarlung Zangbo River in Tibet (Li and Liu, 2009).

GRC-1/3 Glenn Research Center. GRC-1 and −3 are lunar soil simulants that are named after their developer institution- Glen Research Center in USA. They are reported to be made using different mixtures of commercial silica sand products and both are reported to be intended for geotechnical testing purposes (Oravec et al., 2010).

KOHLS-1, which is also known as Korea Hanyang Lunar Simulant-1, is a low-Ti mare type lunar soil which was made for mainly geotechnical purposes (Lee et al., 2012).

BP-1 lunar simulant has been developed in Arizona, United States. This simulant, which is a low-Ti mare type, was made using Black Point basalt flow and it is reported as a simulant intended for geotechnical studies (Rahmatian and Metzger, 2010; Stoeser et al., 2010).

CSM-CL is a lunar mare simulant as it was obtained from a highly alkaline basalt originating from a basaltic scoria from Colorado Lava in Colfax County, New Mexico, United States. The name of the CSM-CL simulant is derived from the developer institution and the resource used: Colorado School of Mines-Colorado Lava. This simulant was developed with the intention of being used for geotechnical experiments (Dreyer and Van Susante, 2010).

OPRH 2N/H2W/H3N/H3W Lunar Highland and OPRL 2N/L2W Lunar Mare Simulants have been developed by Off Planet Research LLC. Both groups of simulants are made from combinations of Shawmere anorthosite and basalt from the San Francisco formation. The particle size distribution of these simulants were arranged by crushing the source materials as needed in order to represent the properties of Apollo 17 regolith (Denevi et al., 2020; Off Planet Research, 2021).

ALRS-1, which is Australian Lunar Regolith Simulant-1, was developed by University of New South Wales by using basalt obtained from Kulnura on the Central Coast of New South Wales in Australia. This lunar soil simulant is reported to specifically developed for being employed in experimental studies on regolith sintering (Bonnano and Bernold, 2015).

TRI-1 is the first lunar soil simulant of India developed within the studies carried out in National Institute of Technology of India in Tiruchirappalli. It was reported that this simulant was designed to reflect the properties of lunar soil samples from Apollo-16 landing site (Jayalekshmi et al., 2019; (Sreenivasulu, 2014)).

European Space Agency (ESA) aims to develop the European Lunar Exploration Laboratory (LUNA) in Germany, that will carry out many studies including practices on Lunar surface operations. For this purpose, LUNA testbed has been designed to have two large areas, in which lunar soil simulants has to be used within, for carrying out the lunar surface operation practices. EAC-1 is the primary lunar soil simulant manufactured for this purpose, and the information about this simulant were published in 2017. EAC-1 is basically basanite quarried from Germany and processed to be used as lunar soil simulant (Manicket al., 2018; Nash et al., 2017). EAC-1A is the part of EAC-1 that contains dust particles and is aimed to be used in the second area of LUNA testbed with the purpose of providing facility to carry out tests in more challenging and realistic scenarios. EAC-1A is classified as "silty sand" and infor-

mation about this simulant was published in 2020 (Engelschiøn et al., 2020).

CLDS-i is the Chinese Lunar Dust Simulant made by the Institute of Geochemistry Chinese Academy of Sciences. It was reported that the source material of this simulant is from volcanic field in northeastern China and that this dust simulant was made by further processing CLRS-1 lunar soil simulant. CLDS-i is proposed to be used for studies like space suit testing and rover (Tang et al., 2017).

BHLD20 (Beihang University Highland Lunar Dust simulant) is another Chinese lunar dust simulant produced in China. Basaltic scoria and anorthositic plagioclase were used as the source material and then the material combination was ground in order to obtain the desired particle size distribution. It is reported that BHLD dust simulant reflects the properties of lunar dust samples to a good extent (Sun et al., 2017).

NEU-1a and b are Chinese lunar soil simulants that were named after their developer institution North Eastern University (NEU) in China. NEU-1a was designed to mimic low-Ti mare lunar soil, where NEU-1b was a designed to have high-Ti content. The source material for both simulants is basalt and volcanic cinder from Huinan County in China. It is reported that the grain compositions of both simulants comply with the real lunar regolith compositional (Li et al., 2019; Li, 2016; Liu et al., 2017).

KLS-1 which is also known as Korea Lunar Simulant-1 was made by using basalt obtained from Cheorwon County of Korea. Lunar soil 14,163 was reported to to be taken as the reference in the production of this simulant and studies carried out yielded the results that the mechanical properties exhibited by KLS-1 was consistent with the targeted regolith sample (Ryu et al., 2018).

DNA-1 was developed as a low-Ti mare lunar soil simulant with the intention to be employed in 3D printing experiments with regolith. The source material was obtained from Bolsena Crater in Italy. DNA-1 was designed as a low-cost alternative to well-known JSA-1 simulant. DNA-1A was developed later on using the same source material used for DNA-1 (Cesaretti et al., 2014; Marzulli and Cafaro, 2019).

TUBS-M and TUBS-T are two "base" simulants made in Germany with the aim to mimic lunar mare and highland regoliths, respectively. The former simulant is made using German alkali-olivine basalt and the latter is made of a Scandinavia metamorphosed gabbroic complex. With these simulants, it is aimed to provide base materials that will serve to produce other specific mare and highland simulants that may be desired only by adding some extra components as it would be needed (Linke et al., 2018; Linke et al., 2020).

LHS-1 and LMS-1 are two lunar soil simulants developed in University of Central Florida by the CLASS Exolith Lab, in USA. The former simulant aims to mimic lunar regolith of average highland areas on the Moon, where the latter is designed to mimic lunar mare soils. These mineral-based lunar soil simulants are reported to reflect the lunar soil texture as they contain mineral and rock fragments in accurate proportions (Cannon and Britt, 2019; Britt and Cannon, 2020).

UoM-B and UoM-W are two low-fidelity lunar regolith simulants that are developed by University of Manchester in UK. These simulants are basically obtained from industrial materials and are reported to contain "volcanic black dust" and "vitreous dust". Hence, they are named after the developer university and their colors, such as University of Manchester-Black (UoM-B) and University of Manchester-White (UoM-W), respectively. The aim of the developers was to provide a low-cost simulant that could be used in large quantities for engineering experiments such as surface excavation and the chemical composition similarity with lunar regolith was not necessarily sought after. It is reported that both UoM-B and UoM-W exhibit relatively similar geotechnical properties as specifically produced lunar simulants have and hence, they are considered as acceptable low-cost simulants for engineering experiments (Just et al., 2020).

LSS-ISAC-1is a recent lunar soil simulant that is made using natural anorthosite rock fragments from Sittampundi, India. This simulant is made mimic lunar highland soil and its chemistry, mineralogy as well as its geotechnical properties are reported to match the Apollo 16 regolith samples. LSS-ISAC-1 is expected to be effectively used in mobility studies, road works, Launchpad and habitation construction studies (Venugopal et al., 2020). LSS-ISAC-1 simulant is designed to carry out investigations to support the Chandrayaan lunar mission that is to be delivered by the Indian Space Research Organization. This mission is intended to have the destination of lunar South Pole, which has the a larger lunar surface area that remains in the shadow compared to the lunar North Pole. The Chandrayaan lunar mission aims to study the lunar topography as well as the elemental abundance; hence, detailed investigations have been carried out and reported on the engineering properties of LSS-ISAC-1 simulant, such as its shear strength and compressibility, as these poperties are expected to affect the lander and rover performances (Prabu et al., 2021). Besides the chemical analysis results showing that the major oxides' percentages are within the range of Apollo 16 Lunar regolith samples, studies also showed that the grain size distribution of LSS-ISAC-1 replicate the grain size of Lunar highland regolith for the range from 30 μm to 1000 μm (Anbazhagan et al., 2021). Additionally, interesting experimental studies have been carried out recently on the dynamic properties of this simulant under simulated moonquake conditions and the detailed results have been reported in the literature (Prabu et al., 2022).

In addition to the aforementioned lunar soil simulants, the results of the studies on the production of a new lunar soil simulant in Turkey have also been recently presented in the literature (Toklu et al., 2022). Nine soil samples obtained from 9 volcanic areas of Turkey together with

ilmenite, olivine, two types of basalts and fly ash samples were used for the production of this simulant named TBG-1 following an optimization process (Toklu et al., 2017). An image of this recently developed lunar soil simulant presented in Fig. 3. With TBG-1, that is reported to be resembling to the Apollo-14 lunar regolith samples, Turkey has become the 10th country in the world producing lunar soil simulants (Toklu et al., 2022).

All aforementioned lunar soil simulants are reported to be used for varying types of lunar investigations (e.g. drilling; landing and surface mobility; production and testing of in-situ construction materials) requiring different mineralogical and chemical properties and varying particle characteristics of lunar soil simulants employed in the experiments. Certainly, each simulant's production/preparation procedure has a direct effect on their properties. It is also known that depending on the type of the tests that a particular simulant was originally designed to be employed in, sometimes the fidelity of the simulants regarding the chemical or physical properties of the regolith they are intended to simulate could be disregarded to an extent during the simulant's production. In fact, Taylor et al. (2016) states that a simulant developed for one kind of testing purpose might totally be performing unsatisfactorily for another category of testing (Taylor et al., 2016).

Research works focused on the manufacture and the characteristics of each one these 54 simulants individually are mentioned to an extent within this section of this work; mainly to allow the readers to know about the existence of all these listed 54 lunar simulants, to provide an *overview* on their origins and types, as well as to provide information on the original references mentioning these lunar simulants for the first time. Further studies detailing other properties and performance of these simulants under certain conditions are also known to be available in the related literature, along with some other valuable studies providing certain simulants' information in a comparative manner. One of these interesting studies is presented by Thannasi et al. in 2021, that provided significant information on the chemical composition, mineralogy, geotechnical and geo-mechanical properties of four mare type and two high-land type lunar soil simulants in a systematical and comparative manner (Thannasi et al., 2021). Such detailed comparative studies on all of the lunar simulants listed in Table 3 are in fact essential in order to provide a better understanding of the characteristics and performance of varying simulants, for guiding researchers to select a more adequate type of simulant for their particular research and also for enabling researchers to design and produce further new lunar simulants exhibiting even higher fidelities.

Fig. 3. TBG-1 lunar simulant developed by using soil samples obtained from 9 volcanic areas of Turkey together with ilmenite, olivine, two types of basalts and fly ash samples (Toklu et al., 2022).

Besides the *overviews* provided in sections 2 and 3 that present respectively the fundamental information regarding lunar missions together with lunar soils, and also the detailed list of all lunar soil simulants that is presented for the first time in this paper in an updated and complete manner, this paper also aims to provide and *overview* on the experimental studies regarding the manufacture of lunar construction materials using lunar soil simulants. The types of these lunar simulant-based construction materials, their manufacture processes, their performance as well as their limitations are described in the following section.

4. Lunar construction materials

With the beginning of outer space missions from 1960's, the idea of colonization of Moon has become a great interest for all leading countries of the world, as this would imply a highly respected technological advancement status as well as yielding access to use resources and other advantages possessed by the Moon. Moon has been always considered as a first step and as a station for pursuing further scientific investigations in deeper parts of the space. In order to be able to establish a lunar outpost, certain facilities should be set up on the Moon such as landing sites and other buildings to provide safe storage and utilization of needed laboratory/investigation equipment as well as habitation of researchers on missions.

As the voyage to the Moon still have limitations, carrying the construction materials from Earth for enabling the construction of Lunar buildings is both complicated and certainly costly (Isachenkov et al., 2021; Oh et al., 2021). Consequently, the concept of in-situ resources utilization (ISRU) has become a core concept for authorities and researchers in planning future lunar missions with lunar settlement (Lim et al., 2017; Toklu, 2001; Toklu and Jarvstrat, 2003). A *qualitative* comparison of timely percentages of in-situ materials and materials brought from the Earth to be used in lunar construction industry can be depicted as in Fig. 4 (Toklu, 2000); illustrating that the quantity of materials brought from the Earth are expected to be used relatively more in the first stages, while the quantity of in-situ materials to be used are expected to

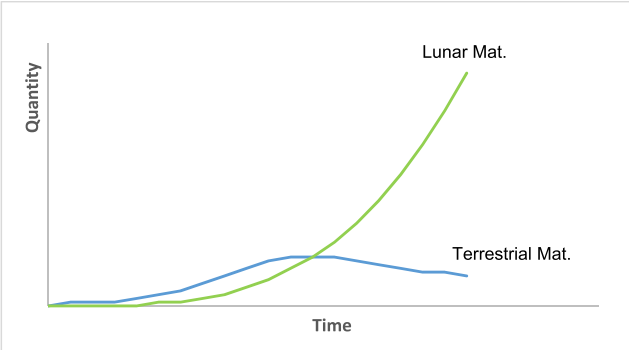

Fig. 4. *Qualitative* comparison of the expected evolution with time of the use of Lunar in-situ and Terrestrial Resources in lunar constructions (Toklu, 2000).

increase in time relatively, for the future stages. The primary local resource considered to be used in making lunar construction materials has been defined as lunar regolith which includes lunar soil of varying particle sizes formed by the impact of meteorites.

First ideas proposed for the construction materials to be used on Moon have been highly influenced by conventional Earth constructions. Table 4 provides a summary of the categories of lunar construction materials mentioned in the related literature, their reported strength performance and some example previous research carried out on these materials.

Compressive and flexural strength values in Table 4 are mentioned in order to provide a general sense and information to the readers on the maximum performance that could be obtained in the fields of these proposed typed of lunar construction materials. Especially the compressive strength value of construction materials is conventionally used in the field of civil engineering in order to define the quality and the performance grade of the materials of concern (Akpinar and Khashman, 2017; Khashman and Akpinar, 2017). However, one should keep in mind that the strength values presented in Table 4 are only for general information purposes and could not be used for direct comparison purposes, since the geometries and the dimension of each tested sample types employed in the mentioned studies differ greatly, as well as the testing setups reported to be chosen in order to meet their individual testing requirements. More detailed information on these mentioned lunar construction materials and the studies proposing them are provided below.

4.1. Regolith-based ordinary cements and concretes

Producing lunar regolith cements and concretes have been an important focus of the related studies since 1980′s. The notable availability of CaO in lunar regolith taken into consideration by Lin (1985) supported the possibility of using regolith concrete as lunar construction (Ponnada and Singuru, 2014). Additionally, other previous studies also show that the chemical composition of lunar regolith and terrestrial fly ash that is a frequently used cementitious material are comparable to a great extent (Khitab et al., 2016). Ponnada and Singuru (2014) describes the option of producing lunar cement by treating lunar regolith at 2000 °C using solar panels.

In another interesting study (Qiao et al., 1990), explain in detail their proposed approach for preparing cementitious materials using lunar soil simulant JSC-1 together with a nano interphase, treated at 260 °C for a period of six hours. Besides being the potential raw material for producing lunar cement, lunar regolith has also been regarded as a very convenient material to serve as aggregates within the concrete, which traditionally form 75% of concrete volume. The specific gravity values determined for lunar regolith of required particle sizes have been observed to be greater than 2.6, which makes lunar aggregates satisfactory for concrete use (Khitab et al., 2016; Ponnada and Singuru, 2014).

As the exposure conditions of the Moon are much more severe than conditions of Earth, manufacture of lunar concrete in the conventional terrestrial manner has certain complications. Lack of liquid water as an in-situ resource to facilitate the hydration of cementitious materials is one of the most important challenges for regular concrete production on Moon (Casanova, 1997; Lim et al., 2017; Omar, 1993; Toutanji et al., 2010). Even if water could be brought from Earth with all its complications, the extreme temperature change experienced on Moon surface which goes beyond + 120 °C, would cause evaporation of water especially when the concrete is in its fresh state where the hydration reaction still continues. Another disadvantageous exposure characteristic of Moon is its severe vacuum condition. Results of previous studies indicate that free water that is needed to be present the concrete mix during hydration process could be lost due to vacuum conditions. Hence, with the absence of water, strength gaining process of concrete could not be completed as desired (Khitab et al., 2016). Ponnada and Singuru (2014) also states that vacuum conditions of Moon are highly likely to yield severely increased porosity of concrete due to loss of water prior to concrete hardening. Increased porosity in hardened concrete implies increased fluid permeability, reduced density, reduced compressive and tensile strengths and reduced radiation protection.

4.2. Sulfur-based lunar concretes

The complications regarding water usage in lunar concrete lead the researchers to consider water-less concrete possibilities. Leonard and Johnson (1988) are known to be the first researchers to propose sulfur-based concrete as a water-less lunar construction material (Ponnada and Singuru, 2014). Sulfur-based concrete had been adopted by engineers for replacing Portland cement concretes to be exposed to harsh terrestrial environments with acid and salt attacks. Besides its water-less character and enhanced durability performance under certain exposure

Table 4
Types of lunar construction materials and their strength performances.

Type of Lunar Construction Material	Examples Studies	Reported Max Compressive Strength	Reported Max Flexural Strength
Ordinary Concrete	Lin, 1985		
Sulfur-based Concrete	Omar, 1993	33.8 MPa & 43 MPa (with fibers)	3.7 MPa
	Toutanji et al., 2010	22 MPa	
	Toutanji et al., 2012	31 MPa	
	Grugel, 2012	~47 MPa	
Polymer-based Concrete	Sik Lee et al., 2015	12.9 MPa	
	Chen et al., 2018		~60 MPa
	Su et al., 2019		47.8 MPa
	(Oh et al., 2021)		16 MPa
Geopolymer Concrete	Wang et al., 2016	35 MPa	
	Wang et al., 2017	~50 MPa	
	Davis et al., 2017	~19 MPa with ambient curing & ~3MPa curing under simulated lunar exposure conditions	
3D Printed Construction Elements/materials	(Cesaretti, 2012)		
	Cesaretti et al., 2014	20.35 MPa	7.1 MPa
	Balla et al., 2012		
	Sitta & Lavagna, 2018	22.7 MPa	
	Meurisse et al., 2018	~5MPa	
	Taylor et al., 2018	19 MPa	
Other sintered & cast lunar regolith blocks			
	Happel, 1993Indyk &	538 MPa	34.5 MPa
	Benaroya, 2017	218.8 MPa	
	Mueller, 2017 (Based on NASA studies)	>200 MPa	
	Kim et al., 2021	37 MPa	
Recent Alternative Composite Lunar Construction Materials:			
• Regolith-based magnesium oxychloride composites with graphene	(Lauermannová et al., 2021)	68.2 MPa with laboratory curing 66.7 MPa cyclic heating &freezing	19.3 MPa 18.3 MPa
• Regolith Bricks with microbial induced calcite precipitation & biopolymer	Dikshit et al., 2021	2.38 MPa	
• Biopolymer-Bound Regolith Composites	Rosa et al., 2021	25 MPa	

conditions, sulfur-based concrete is also known for its accelerated hardening and strength gaining. It is known to reach 70–80% of its ultimate strength only within 24 h (Casanova, 1997).

Sulfur-based concrete includes a mixture of sulfur and aggregates only and these two constituents do not chemically react with each other while forming the construction material. In this mixture, sulfur alone acts as the binder keeping the aggregates together, replacing cement and water in conventional concrete that undergo hydration reaction resulting in sticking, binding and strength gaining characteristics in concrete. With its thermoplastic character, sulfur is first melted at around 119 °C and as the liquid surrounding the aggregates, it undergoes a phase change process and hardens at around 148 °C while binding the aggregates and forming the hardened sulfur-based concrete (Toutanji et al., 2010). Sulfur is known to be present within the resources of Moon in the form troilite FeS mineral and its acquisition in the elemental form troilite is reported to be possible at 1000–1200 °C that is achievable with standard solar concentrators (Casanova, 1997).

Several researchers carried out noteworthy studies on the performance and characteristics of sulfur-based concrete as a potential lunar construction material. In 1993, Omar (1993) carried out experimental studies on manufacturing concrete mixes made of JSC-1 lunar simulant with systematically increasing sulfur content. Results of this study indicated that increasing sulfur addition yielded increased compressive and tensile strength values, till the optimum sulfur content, which was 35% in their case, was reached with a resultant 33.8 MPa concrete compressive strength. Another interesting result reported by (Omar, 1993) indicated that the sulfur-concrete workability was unacceptably low for low sulfur contents such as 30%. It was observed that increasing the sulfur content yielded faster setting time for the concrete mix. The study also reported 2200 kg/m^3 density for the sulfur-based concrete mix, which is comparable to the densities of conventional

concretes with no sulfur content. An additional aspect of this study covered the inclusion of fibers in the mix with the intention of improving strength performance. Results showed that with 2% metal fiber inclusion, the compressive strength could be increased beyond 40 MPa for the mixes having 35% sulfur content.

A higher compressive strength value has been reported by for the case of sulfur-based concrete prepared without any fiber addition. Toutanji et al. (2012) presented experimental results for sulfur-based concrete samples prepared with 35% JSC-1 lunar soil simulant and 65% sulfur that yielded 31 MPa compressive strength. The study also reported noteworthy remarks on impact test and radiation investigations of the produced sulfur-based concrete samples as well, which are significant considering the conditions that lunar constructions would be exposed to.

Even though sulfur-based concrete has been reported to exhibit satisfactory strength performance, previous studies in the literature also reported certain noteworthy drawbacks of this material for lunar exposure conditions. It is expected that sulfur concrete could deform and sulfur would melt under lunar temperature conditions that could rise up to +128 °C, which is higher than the melting temperature of sulfur (Gracia and Casanova, 1998; Meyers and Toutanji, 2007). As the temperature on the Moon equator varies between +128 and −180 °C in a cyclic manner, investigations on freeze–thaw durability performance of sulfur concrete are also significant. The experiments carried out by American Concrete Institute in 1988 according to ASTM C666 Method A, with 300 cycles yielded retention of the dynamic modulus of elasticity of sulfur-based concrete up to around 60% (Casanova, 1997). In another study, (Grugel, 2012) investigated the effect of strength loss in sulfur concrete due to exposure to temperature cycles ranging between 20 °C and −191 °C. It was reported that 80 cycles of temperature change with the defined range drastically decreased the compressive strength of sulfur-based concrete samples from 35 MPa to 7 MPa (Grugel, 2012). Besides disadvantageous temperature conditions, the vacuum conditions on the Moon are also expected to have a degrading effect on the sulfur-based concrete as well ()(Wan et al., 2016). In their study, Grugel and Toutanji (2008) have investigated the quantity of sublimation to be experienced by sulfur concrete samples subjected to a vacuum environment with 5×10^{-7} torr, at 20 °C for 60 days by considering their weight losses. It was also predicted that the sublimation rates would be significantly increased at higher temperatures such as 120 °C (Grugel and Toutanji, 2008).

4.3. Polymer-based lunar concretes

In the pursuit of water-less lunar concrete, other researchers investigated the feasibility of using lunar regolith with different binders such as polymers (Chen et al., 2018; Koh et al., 2010; Oh et al., 2021; Lee et al., 2015; Su et al., 2019). Sik Lee et al. (2015) prepared polymer concrete by mixing grounded basalt for mimicking lunar regolith, together with polyethylene forming the 10% of the concrete mixture. These samples were reported to gain a maximum compressive strength value of 12.9 MPa in a short time such as 5 hours (Sik Lee et al., 2015). In another study, Chen et al. (2018) presented the experimental results of the performance of lunar simulant-based composites prepared with a low quantity of epoxy as the polymer binder. 2–8% of epoxy was mixed with JSC-1A lunar soil simulant and the samples were compacted as a part of the preparation procedure. Samples were tested under flexure and it was reported that flexural strength increased with the increasing epoxy content and 2–5% epoxy was sufficient to manufacture samples that had comparable and higher flexural strength performance than typical steel reinforced concrete samples (Chen et al., 2018). Su et al. (2019) studied the fatigue behavior of samples prepared with JSC-1A lunar soil simulant mixed with 4% epoxy. Su et al (2019) reported that the samples exhibited ductile-like behavior. It was also reported that the observed S-N data for fatigue and fatigue performance of these samples were superior when compared with the performance of typical steel-reinforced concrete under the same applied loading. Oh et al. (2021) also prepared samples with JSC-1A simulant mixed with 4% epoxy. It was reported that the flexural strength of the samples was ranging between 12 and 16 MPa, which is defined to be in a comparable range with steel-reinforced concrete. According to Oh et al. (2021), using samples of lunar regolith mixed with epoxy would be a fast and simple approach for ISRU in manufacturing structural elements on Moon.

4.4. Geopolymerized regolith concretes

Some other researchers investigated the feasibility of using geopolymerized lunar regolith to produce geopolymer concrete-like construction materials in order to be employed in lunar constructions. Geopolymers are known to be formed through dehydration-polycondensation process and even though some water is present in the activator solutions, the ultimate geopolymer product does not include water in its structure (Wang et al., 2017). Wang et al. proposed the use of an efficient water recycling device included in the manufacture procedure of the depolymerized lunar regolith samples, since it could provide the benefit of almost zero water consumption during the manufacture (Wang et al., 2016; Wang et al., 2017). Wang et al. (2016) presented the experimental results on the samples prepared with tektites used as simulants mimicking lunar regolith activated with 5 M sodium hydroxide and sodium silicate solutions serving as alkaline activators in two separate cases for yielding geopolymerization. The samples made with these two activators were reported to have over 26 and 35 MPa compressive strengths, respectively. 30 freeze–thaw cycles applied to the both type of samples were reported to cause nearly zero and around 10% strength losses, respectively, for their two types of

samples. In 2017, Wang et al.(2017) carried out similar experiments using volcanic ash as the simulant mimicking lunar regolith activated with sodium sulfate solution and exposed the samples to vacuum conditions and 30 freeze–thaw cycles. The samples exposed to these conditions were reported to yield around 45 MPa compressive strength values, while 98.6% of the water used during manufacture could be successfully recycled (Wang et al., 2017). Even though Wang et al. (2017) could report high compressive strength values for the geopolymerized lunar simulant samples, Davis et al. (2017) reported severe strength decrease (yielding strength values as low as ~3MPa) for samples in their study, caused by exposure conditions including vacuum and average daytime lunar heat (i.e. 106 °C). The study presented by Davis et al. (2017) included geopolymer samples prepared with JSC-1A lunar simulant activated by sodium hydroxide and sodium silicate solutions. It is expected that the type of the simulant used to mimic the lunar regolith, varying geopolymerization procedures, as well as varying test conditions attempting to simulate lunar exposure conditions yielded the observed variations in the determined performance results. Hence, careful and somehow harmonized future studies on geopolymerized lunar regolith samples are needed to further understand the feasibility of employing this material effectively in lunar constructions.

4.5. Lunar construction materials with 3D printing

Concrete-like lunar construction materials have been a great interest of researchers working in this field mainly because of the extensive familiarity and the solid technical background gained for both the manufacture and the performance requirements of conventional terrestrial concrete. However, continuously developing technologies, as well as emerging perspectives and understandings on the local conditions, challenges and needs for extra-terrestrial settlements have also yielded different options and approaches for manufacturing lunar construction materials. Using additive manufacturing technologies (AM), which is also known as 3D Printing, is regarded as the key approach by the researchers working in the field of Space Architecture for constructing different habitation facilities on Moon with in-situ resource utilization. With the additive manufacturing approach, construction elements even having complex geometries could be manufactured by joining materials layer upon layer using 3D model data using technique (Lim et al., 2017). The AM technology is considered as highly preferable in this field, since a minimum level of preprocessing is required for the raw materials to be used and also as it is known to be a highly versatile technology enabling researchers to manufacture numerous different types of components by making just minor reconfigurations (Isachenkov et al., 2021).

Within a project funded by European Space Agency, Cesaretti et al. (2014) presented results of investigations on manufacturing lunar building block and structure components made of lunar soil simulant named DNA-1 using a special 3D printing technology (Cesaretti, 2012; Cesaretti et al., 2014). In these works, granular simulant material mixed with pulverized metal oxide was sprayed with a liquid binder and then the second layer of simulant was laid on top. While a chemical reaction was occurring between the metal oxide in both top and bottom layers of simulant and the liquid binder in between, the layers were also compressed. Cesaretti et al. (2014) reported successful application of this technique to manufacture a real size 3D printed component of a designed outpost. The compressive and flexural strength performance of 3D-printed regolith blocks with 13% porosity were reported to be 20.35 and 7.11 MPa, respectively (Cesaretti et al., 2014).

It is known that evaporation, melting and sublimation issues in previously mentioned lunar construction materials may occur as discussed earlier due to the high values of the harsh cyclic temperature change range of Moon. Moreover, the low values that are experienced in the Moon's temperature change cycle may cause freezing issues in the component manufacture case of Cesaretti et al. (2014) which uses a liquid binder, if adequate measure are not taken carefully. Hence, new approaches have been sought after by researchers in order to overcome Moon's exposure condition challenges while manufacturing construction materials with lunar regolith. Using minimum amount of liquid binders under these exposure conditions and applying minor pre-treatment to the regolith have been regarded beneficial approaches for acquiring further success in lunar construction material manufacture. Hence, *sintering-based* additive manufacturing for lunar constructions have become significant for research studies in this field (Lim et al., 2017).

With sintering process, compacted solid materials are acquired with a heat treatment at a temperature lower than the melting point of the material, allowing the particles to come closer and bond at new positions and hence yielding a reduction in the pore volume of the resultant material. A noteworthy study carried out by Taylor et al. (2018) presents the experimental results on the manufacture and the performance of micro-trusses made with sintering-based 3D-printing of liquid inks containing JSC-1A lunar soil simulant, poly (lactic-coglycolic acid) copolymer and organic diluents. The components (i.e. micro-trusses) were prepared with direct-extrusion 3D-printing of this liquid ink and then they were sintered with appropriate furnaces at air or hydrogen atmospheres. The compressive strength performance of the manufactured components varied between 1 and 19 MPa depending on sintering duration, sintering temperature and the atmosphere that they were exposed to (Taylor et al., 2018).

Sintering lunar soil simulants by laser, solar concentrator or microwave energies are considered to be feasible for manufacturing lunar construction elements. Balla et al. (2012) presented the result of an experimental study on laser sintering JSC-1AC lunar soil simulant. Results showed that samples could successfully be produced with

high geometrical accuracy. In another interesting study, Sitta and Lavagna (2018) produced small 3D-printed components made of NU-LHT-2 M lunar highland simulants by employing selective laser applications and hence, 22.7 MPa compressive strength was reported for the manufactured samples. Another more recent study in the field of additive manufacturing also presented interesting experimental results based on the same simulant, NU-LHT-2 M, mainly on its processability by Laser Powder Bed Fusion. Multi-layer cubical samples were prepared and their mechanical properties were then tested. Result of this study showed that these samples could attain compressive yield stress values of 31.4 MPa (Caprio et al., 2020).

On the other hand, some other researchers support the idea that using only laser sintering is not as efficient as desired, since the quantity of energy needed to apply laser sintering on a real size lunar construction would be extremely high. Also, it is expected that a long duration of time would be required in order to complete printing process, since laser sintering could be applied to treat a small volume of material at a unit time (Lim et al., 2017). Some other researchers promoted the use of solar sintering due to its advantages such as solar energy is readily and limitlessly available on Moon. In their study, Meurisse et al. (2018) presented the experimental results on the feasibility of using solar sintering method for 3D printed lunar bricks made of JSC-1A and JSC-2A lunar soil simulants. The authors reported that the manufactured bricks yielded 5 MPa compressive strength and proposed ideas to further improve the manufacture technique (Meurisse et al., 2018) Solar sintering additive manufacturing is also reported to have certain drawbacks such as the need to clean the solar concentrator's glasses from lunar dusts and the fact that the solar concentrator's efficiency would be affected by the optical properties of lunar regolith that varies within mare and highland regions of the Moon. It was also reported that solar sintering could treat a limited depth of material when compared to microwave sintering (Lim et al., 2017). Microwave energy-based sintering could yield superior results as wider areas of regolith could be treated in this way. The natural chemical composition of lunar regolith is reported to facilitate efficient microwave sintering (Srivastava et al., 2016) and hence, successful microwave sintering 3D printing for the manufacture of lunar construction elements could become possible, even though further investigations on the determination of most effective microwave frequency are reported as being still necessary (Lim et al., 2017).

Being a relatively recent technological development, 3D printing is considered to have a high potential that could transform ISRU capabilities; consequently, enhancing the level of success of varying lunar missions (Ellery, 2020).

4.6. Other sintered or cast regolith blocks

Even though sintering-based lunar construction material manufacture processes in general should be still improved in order to achieve the exact product performance desired on real lunar applications, sintered lunar construction materials having promising engineering performances have been reported in the previous literature. For instance, Indyk and Benaroya (Indyk and Benaroya, 2017) sintered lunar soil simulant JSC-1A in two batches as having 1.44% and 11.78% porosities and reported remarkably high compressive strength values for these batches as 218.8 MPa and 84.6 MPa, respectively. A recent study using microwave sintering process, reports compressive strength values such as 37 MPa for samples made with KLS-1 lunar simulant (Kim et al., 2021). The authors of this study also reported that increasing microwave temperature lead to increasing densities and increasing compressive strength values for the tested samples (Kim et al., 2021).

In another study, Happel (1993) reported the strength performance of cast regolith (cast basalt) samples that were prepared by melting the regolith, later molding and allowing it to cool slowly. Happel (1993) reported that cast basalt prepared in this manner yields a brittle behavior even though it exhibits a high strength performance, such as 538 MPa and 34.5 MPa of compressive and tensile strengths respectively. Also, Mueller (2017) presented results on research studies carried out within NASA, indicating more than 200 MPa compressive strength for sintered basalt regolith samples. All these research results indicate that sintered lunar construction materials could potentially serve successfully in making the future lunar habitats and infrastructures.

On the other hand, some studies suggest that using cast basalt will be only serving to bear compression, and in order to construct tension-bearing structures, materials such as regolith fibers having high tensile strength should be employed (Farries et al., 2021).

Mentioned studies show that the effects of Moon's harsh exposure conditions should definitely be taken into consideration for selecting the appropriate construction material type to be used. In addition to the lunar environmental conditions to be considered, the type of the lunar construction, its design criteria, needs and parameters are also critical in achieving success in the task of reliable lunar outpost constructions (Järvstråt and Toklu, 2004; Toklu, 2000a, Toklu, 2000b; 2005; Yerel and Toklu, 2014). Therefore, adequate lunar construction material should be defined and selected considering the specific performance needs of designed lunar buildings as well.

4.7. Recent alternative composite lunar construction materials

4.7.1. Regolith bricks with microbial induced calcite precipitation

In addition to more conventional concrete-like lunar construction materials that have been suggested for some decades, some recent and innovative studies suggest the use of Microbial Induced Calcite Precipitation (MICP) in order to make bricks with lunar soil. In their interesting

study, Dikshit et al. (Dikshit et al., 2021), presented experimental results for space bricks manufactured by using Indian lunar soil simulants (LSS) together with Sporosarcina pasteurii which was used as the bacterial strain to initiate the microbial induced calcite precipitation. Additional batches were prepared with the inclusion of guar gum, which is a natural biopolymer, and glass fibers used for structural reinforcement. The bricks made with additional guar gum yielded the highest compressive strength which was reported as 2.38 MPa, where the glass fiber addition was reported by the researchers to yield a reduction in the compressive strength of the bricks fabricated.

4.7.2. Regolith-based magnesium oxychloride composites with graphene

Another recent interesting study by Lauermannova et al. (2021) proposes the use of a special composite material for lunar buildings. This composite material is proposed to be prepared in a fresh sate by including lunar regolith, MgCl2ṡ6H2O, MgO and also graphene nanoplatelets added to improve the material properties and then casted into selected molds to allow hardening. In their study, Lauermannova et al. prepared samples of this described regolith-based magnesium oxychloride composite with graphene addition by employing lunar soil simulant OPRL2N (Lauermannová et al., 2021). Two sets of samples, each consisting a variety of simulant contents, were subjected to laboratory curing and also a designed cyclic heating and freezing exposure conditions were tested. The samples exposed to cyclic heating and freezing cycles were observed slightly lower mechanical properties than the ones cured in the laboratory conditions; nevertheless, they exhibited remarkably high compressive and flexural strength values as around 67 and 18 MPa respectively, for batches including 50 wt% of regolith (i.e. lunar soil simulant).

4.7.3. Biopolymer-bound regolith composites

A new type of biopolymer-bound soil composite (BSC) lunar construction material has been recently proposed by Rosa et al. (2021). This material was manufactured by using "Bovine blood plasma proteins" together with JSC-1A lunar soil simulant. The authors of the mentioned works prepared nineteen batches having different biopolymer to simulant ratios and biopolymer solution concentrations. The highest compressive strength of 25 MPa was reported to be obtained with the batch having 12.4% biopolymer to soil ratio and at 48% biopolymer concentration.

5. Conclusive remarks

The Moon is the closest space object to Earth. It is thus the first object on which there will be the extraterrestrial colonization works. It is evident that construction will form a very important part of these works. That is why the lunar environment, its ground and soil conditions are investigated with all possible means available to mankind. For this purpose, soil samples, not of sufficient quantity until now, have been transported to Earth. In paragraphs above, an overview about the lunar soil samples that are brought to Earth from Lunar missions pioneered by three countries has been given, with an emphasis on their non-uniformity.

In order to have enough material to be used for research purposes; in ten countries, over fifty simulants of real lunar soils are produced until now. Second overview in this paper, which presentes an up-to-date and complete list of these 54 simulant names together for the very first time in the literature, is about the origin and the main properties of these lunar soil simulants. In this study, it is observed that some of these simulants have focused on resembling the chemical composition of lunar regolith to *a possible extent*, while some other are only focused on reflecting only certain physical, mechanical, and geometrical properties, depending on the aimed specific area of testing. Hence, even though there are numerous lunar soil simulants developed in the last decades, there is still work to be done for taking into account both chemical and mechanical properties of lunar soils together, in order to enable the task of carrying out more realistic experiments with more accurate results prior to taking any critical action in a real lunar mission.

As far as constructions are concerned, the next step is finding ways of producing construction materials using lunar soil as the main raw material. As the third objective in this paper, an overview is presented on use of lunar regolith for this purpose. This overwiew have shown that various techniques have been tried in this direction that yielded promising results in some cases and significant insights on each technique's and material type's feasibility; however, none of these studies could be classified as being in a conclusive stage yet.

The study have shown that there is still a long way to go for producing lunar soil simulants with *all* properties matching with the original samples and since the original samples are not uniform, and special care has to be taken for which area the sample is being produced. Further studies on assessing the properties of existing varity of simulants should be carried out in a comparative manner in order to provide insights on the fidelity and performance of simulants, as well as for providing further guidance on the manucfacture of both new simulants and lunar construction materials from them. The study have also shown that the work to be accomplished in producing lunar construction materials out of lunar soils is still at its infancy; further advancements should beattained in order to have comparable and applicable products in this sense. There is still a need for carrying out systematical experimental works for finding the best materials to be used in lunar environment, using lunar soil as the basic component, with little additions brought from the Earth, and using technol-

ogy that can be applied on the Moon. It is certain that the applications on the Moon will pioneer similar works to be performed on other space bodies, starting with the Mars.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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